dence estimates across different levels of abstraction.
Existing Methods: Existing approaches typically focus on flat, single-level uncertainty estimates or simple hierarchical decompositions that don't fully capture the complex, nested nature of semantic understanding.
Motivation: Drawing inspiration from fractal geometry, where patterns repeat at different scales, we propose a method that recursively decomposes concepts and queries into self-similar sub-components, allowing for a more nuanced and scale-invariant approach to uncertainty quantification.
Proposed Method: We present Semantic Fractal Decomposition (SFD), a prompting technique that guides the model to recursively break down a given query or concept into smaller, self-similar components. At each level of decomposition, the model is asked to provide a confidence estimate. The process continues until a predefined depth is reached or the model indicates it can no longer meaningfully decompose the concept. The final uncertainty estimate is then constructed by aggregating these multi-level confidence scores using a novel fractal dimension-inspired algorithm. This approach allows for capturing uncertainty that may be present at different semantic scales and provides a more robust and consistent measure of the model's confidence across varying levels of abstraction.
Experiment Plan: We will evaluate SFD on a diverse set of tasks ranging from simple factual queries to complex, multi-faceted questions in domains like philosophy, science, and law. We will compare its performance against traditional flat confidence estimation techniques and simpler hierarchical methods. Key metrics will include the consistency of uncertainty estimates across related queries at different levels of abstraction, the correlation between fractal-aggregated confidence scores and actual model performance, and the interpretability of the decomposition process.

Similarity: 0.81

I Overlap Between AI Ranking and Expert Reranking

We show the overlap between the `AI Ideas` condition and the `AI Ideas + Human Rerank` conditions in Table 12. We note that 17 out of the 49 ideas in the `AI Ideas + Human Rerank` condition are also ranked as top ideas in the `AI Ideas` condition by the AI ranker, while the other 32 are not.

Topic	Overlap	New
Bias	2	2
Coding	4	5
Safety	2	3
Multilingual	5	5
Factuality	2	9
Math	2	2
Uncertainty	1	5
Total	18	31

Table 12: Overlap of ideas between `AI + Human Rerank` and `AI` conditions, broken down by topic.

J Quality Control of Human Expert Ideas

Each expert is instructed to choose one of the seven specified topics and write one idea on it within 10 days, following the given template in the annotation document. We included an honor code statement to ask the participants to not use any AI tools in their idea writing. We collected $N = 50$ ideas originally and manually checked all of them for quality control. We filtered out one of them as being essentially a paraphrase of an existing paper's abstract. We compensated the participant nevertheless but excluded them from the review task.

K Breakdown of Participant Positions

We show the detailed position breakdown of our 49 idea-writing participants in Table 13 and the positions of our 79 reviewer participants in Table 14.

Position	Count
Postdoc	1
PhD	36
Master	9
Undergraduate	1
Research Scientist	1
Machine Learning Engineer	1

Table 13: Positions of the 49 idea writing participants.

Position	Count
Postdoc	7
PhD	63
Master	5
Research Scientist	3
Machine Learning Engineer	1

Table 14: Positions of the 79 idea reviewing participants.

L Institutions of the Idea Writing Participants

Institution	Count
Stanford University	11
University of Southern California	6
University of Maryland	3
University of Illinois Urbana-Champaign	3
Johns Hopkins University	3
Columbia University	2
Carnegie Mellon University	2
University of Pennsylvania	1
Princeton University	1
Penn State University	1
Portland State University	1
Stony Brook University	1
University of Chicago	1
University of Washington	1
UC Berkeley	1
UCSD	1
Massachusetts Institute of Technology	1
George Washington University	1
Yale University	1
University of Toronto	1
Georgia Institute of Technology	1
National University of Singapore	1
Peking University	1
Tsinghua University	1
LinkedIn	1
Norm AI	1

Table 15: Institutions of the 49 idea writing participants.

M Institutions of the Idea Reviewing Participants

Institution	Count
Stanford University	25
UC Berkeley	4
UT Austin	4
University of Maryland	4
Princeton University	3
University of Washington	3
University of Southern California	3
Carnegie Mellon University	3
University of Chicago	2
Johns Hopkins University	2
UCLA	2
Georgia Institute of Technology	2
University of Illinois Urbana-Champaign	2
Tsinghua University	2
Stony Brook University	1
Ohio State University	1
National University of Singapore	1
University of Michigan	1
Dartmouth College	1
Massachusetts Institute of Technology	1
University of Pennsylvania	1
University of Toronto	1
Portland State University	1
Penn State University	1
New York University	1
Columbia University	1
UC Santa Barbara	1
Brown University	1
Amazon	1
LinkedIn	1
Norm AI	1
AMD	1

Table 16: Institutions of the 79 reviewer participants.